

No 1. EDA dan Modeling

Untuk Classifier, model paling bagus adalah XGBoost.

- F1-score tertinggi sebesar 0.7212. Dimana ini indikator terbaik untuk imbalanced data karena memberikan bobot yang sama untuk setiap kelas.
- XGBoost memiliki F-1 untuk Class 0 sebesar 0.5325, lebih tinggi dibanding LightGBM (0.5176) dan jauh melampaui Random Forest (0.3878).

Selain F-1 score, kita juga melihat recall karena recall memberi tahu seberapa banyak class 0 yang berhasil dideteksi oleh model. Disini Recall XGBoost hampir mendekati LightGBM

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Untuk Regression, model paling bagus adalah LightGBM.

- MAE (1.1030) & RMSE (1.4193) Terendah. LightGBM memiliki nilai error paling kecil. Ini berarti prediksi gaji (LPA) yang dihasilkan oleh LightGBM paling mendekati nilai aslinya dibandingkan Random Forest dan XGBoost.
- R2 Score (0.7845) Tertinggi. Nilai R2 yang paling mendekati 1 menunjukkan bahwa LightGBM mampu menjelaskan variansi data sebesar 78.45%, lebih tinggi daripada dua model lainnya.

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No 2. Membuat Pipeline

Step 4: Model Evaluation

Loading Classification model from MLflow (Run ID: 9621f823fd7f43e4affdf2a7d24f7f73)...

Downloading artifacts: 0% | 0/1 [00:00<?, ?it/s]

Downloading artifacts: 100% | 5/5 [00:00<00:00, 794.98it/s]

Classification Evaluation: Accuracy=0.8700 | F1-Score (Macro)=0.6900

Loading Regression model from MLflow (Run ID: 40ea801f72554428b4d2d8cc6f32651b)...

Downloading artifacts: 0% | 0/1 [00:00<?, ?it/s]

Downloading artifacts: 100% | 5/5 [00:00<00:00, 227.28it/s]

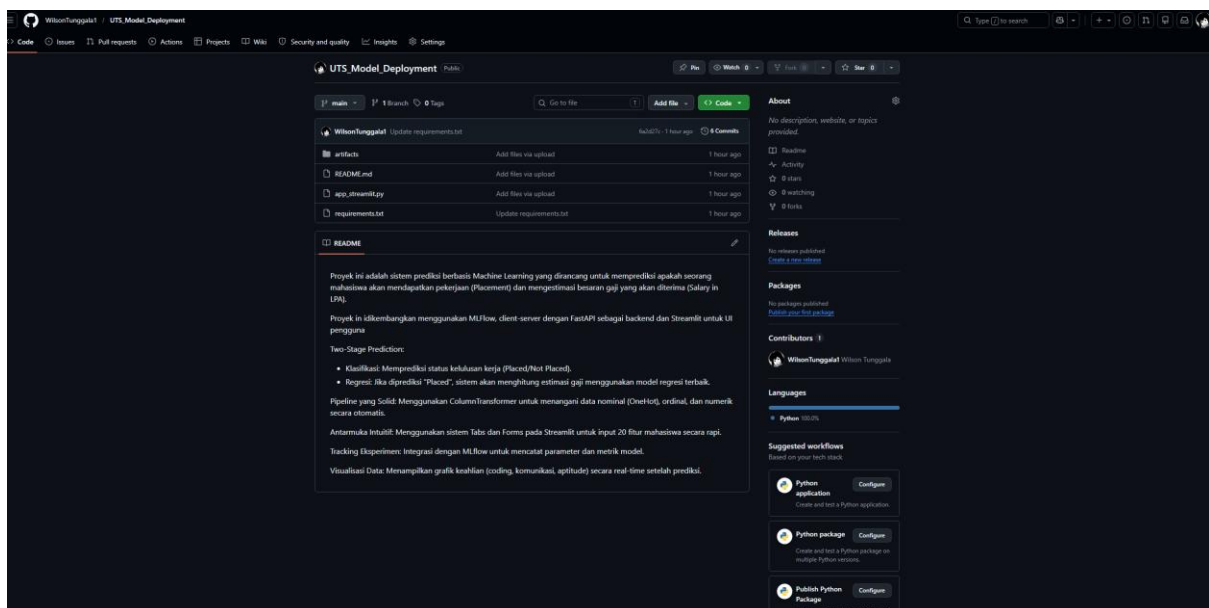
C:\Users\User\miniconda3\envs\model_deployment\Lib\site-packages\sklearn\utils\validation.py:2691: UserWarning: X does not have valid feature names, but LGBMRegressor was fitted with feature names

warnings.warn(
Regression Evaluation: MAE=1.0854 LPA | RMSE=1.3948 LPA | R2=0.7797

Step 5: Deployment Decision

SUCCESS: Both models met the thresholds. Approved for deployment!

No 3. Bangun Aplikasi Web dan Deploy ke Streamlit cloud



Github: https://github.com/WilsonTunggal1/UTS_Model_Deployment/tree/main

App: <https://apppppy-6zxuqoew8rarbwxyjmwwhyw.streamlit.app/>

Informasi Aplikasi

Aplikasi ini menggunakan model Machine Learning Two-Stage:

Tahap 1: LightGBM Classifier untuk memprediksi kelulusan penempatan.

Tahap 2: LightGBM Regressor untuk memprediksi gaji (khusus yang lolos).

Dibuat untuk keperluan MLOps Pipeline.

Aplikasi ini memprediksi apakah seorang mahasiswa akan mendapatkan pekerjaan (Placed) atau tidak. Jika diprediksi mendapatkan pekerjaan, sistem akan secara otomatis memprediksi estimasi gaji (Salary LPA) yang akan didapatkan.

Masukkan Data Mahasiswa

Akademik & Keahlian

CGPA (0.0 - 10.0)

7,50

Pengalaman & Proyek

Persentase Kehadiran (%)

85

Profil Pribadi

Jam Belajar Per Hari

4

Jumlah Backlog (Mata kuliah mengulang)

0

Jurusan (Branch)

CSE

Rating Skill Coding (1-100)

50

Rating Skill Komunikasi (1-100)

50

Rating Skill Aptitude (1-100)

50

Prediksi Sekarang

No 4. FastAPI & Streamlit

default

GET / Read Root

Parameters

No parameters

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/

Server response

Code

Details

200

Response body

{ "message": "Welcome to the Student Placement Prediction API" }

Response headers

content-length: 61
 content-type: application/json
 date: Thu, 23 Apr 2026 14:14:07 GMT
 server: uvicorn

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

"string"

Test Case 1:

Responses

Curl

```
curl -X 'POST' \
'http://127.0.0.1:8000/predict' \
-H 'accept: application/json' \
-H 'Content-Type: application/json' \
-d '{
  "gender": "Male",
  "branch": "IT",
  "part_time_job": "No",
  "internet_access": "Yes",
  "cgpa": 5.5,
  "backlogs": 0,
  "study_hours_per_day": 6,
  "attendance_percentage": 90,
  "projects_completed": 3,
  "internships_completed": 2,
  "coding_skill_rating": 85,
  "communication_skill_rating": 80,
  "aptitude_skill_rating": 75,
  "hackathons_participated": 2,
  "certifications_count": 4,
  "sleep_hours": 7,
  "stress_level": 4,
  "family_income_level": "Medium",
  "city_tier": "Tier 1",
  "extracurricular_involvement": "None"
}'
```

Request URL

http://127.0.0.1:8000/predict

Server response

Code

Details

200

Response body

```
{
  "placement_status": 1,
  "placement_probability": 0.9926356673248662,
  "estimated_salary_lpa": 11.449365949752913
}
```

Response headers

```
content-length: 107
content-type: application/json
date: Thu, 23 Apr 2026 14:17:22 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Test Case 2:

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
'http://127.0.0.1:8000/predict' \
-H 'accept: application/json' \
-H 'Content-Type: application/json' \
-d '{
  "gender": "Female",
  "branch": "CSE",
  "part_time_job": "No",
  "internet_access": "No",
  "cgpa": 5.2,
  "backlogs": 4,
  "study_hours_per_day": 2,
  "attendance_percentage": 65,
  "projects_completed": 0,
  "internships_completed": 0,
  "coding_skill_rating": 25,
  "communication_skill_rating": 40,
  "aptitude_skill_rating": 35,
  "hackathons_participated": 0,
  "certifications_count": 0,
  "sleep_hours": 9,
  "stress_level": 8,
  "family_income_level": "Low",
  "city_tier": "Tier 3",
  "extracurricular_involvement": "None"
}'
```

Request URL

http://127.0.0.1:8000/predict

Server response

Code

Details

200

Response body

```
{
  "placement_status": 1,
  "placement_probability": 0.6908090710639954,
  "estimated_salary_lpa": 8.966212644979212
}
```

Response headers

```
content-length: 106
content-type: application/json
date: Thu, 23 Apr 2026 14:18:01 GMT
server: uvicorn
```

Responses

Code	Description	Links
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Test Case3:

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/predict' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "gender": "Male",
    "branch": "",
    "part_time_job": "No",
    "internet_access": "Yes",
    "cgpa": 7.2,
    "backlogs": 0,
    "study_hours_per_day": 4,
    "attendance_percentage": 82,
    "projects_completed": 1,
    "internships_completed": 1,
    "coding_skill_rating": 55,
    "communication_skill_rating": 65,
    "aptitude_skill_rating": 60,
    "hackathons_participated": 0,
    "certifications_count": 2,
    "sleep_hours": 6,
    "stress_level": 5,
    "family_income_level": "Medium",
    "city_tier": "Tier 2",
    "extracurricular_involvement": "Medium"
  }'
```

Request URL

http://127.0.0.1:8000/predict

Server response

Code

Details

200

Response body

```
{
  "placement_status": 1,
  "placement_probability": 0.9985187649726868,
  "estimated_salary_lpa": 11.587178645768919
}
```

Response headers

```
content-length: 187
content-type: application/json
date: Thu, 23 Apr 2026 14:18:40 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Prediction	View Details

Test Case Streamlit

Masukkan Data Mahasiswa

Akademik & Keahlian

Pengalaman & Proyek

Profil Pribadi

CGPA (0.0 - 10.0)

7,00

Rating Skill Coding (1-100)

77

Persentase Kehadiran (%)

100

Rating Skill Komunikasi (1-100)

60

Jam Belajar Per Hari

7

Rating Skill Aptitude (1-100)

60

Jumlah Backlog

5

Jurusan (Branch)

IT

Prediksi via API

Hasil Prediksi

Status: PLACED (Diterima Bekerja)

Estimasi Gaji (Salary):

Tingkat Kepercayaan Model: 72.93%

Lakhs Per Annum (LPA)

16.07 LPA

Analisis Profil Keahlian Anda:

Masukkan Data Mahasiswa

Akademik & Keahlian **Pengalaman & Proyek** **Profil Pribadi**

CGPA (0.0 - 10.0) 0,00	Rating Skill Coding (1-100) 55
Persentase Kehadiran (%) 42	Rating Skill Komunikasi (1-100) 60
Jam Belajar Per Hari 6	Rating Skill Aptitude (1-100) 24
Jumlah Backlog 3	
Jurusan (Branch) IT	
Prediksi via API	

Hasil Prediksi

Status: NOT PLACED (Belum Mendapatkan Pekerjaan)

Estimasi Gaji (Salary):

Tingkat Kepercayaan Model: 37.22%

Lakhs Per Annum (LPA)

0.00 LPA